

Reflection

Part 1: Concepts

1. Explain the difference between classification, detection, and segmentation in your own words. Give one workforce example of each.

Classification is identifying what an object is not where it is located. Detection takes it a step further by outlining objects with boxes. Segmentation is even further by outlining objects specifically around their edges, instead of a box with other background noise. In a workplace environment, classification can be used to identify PPE items present in an image. Detection would identify each worker individually and the PPE they are wearing. Segmentation could be used to outline exact PPE or the worker and their PPE with more precise measurements when needed.

2. What is Non-Maximum Suppression, and why does YOLO use it?

Non-Maximum Suppression basically removes duplicate boxes, around the same object. Sometimes YOLO put multiple boxes around objects with different confidence scores. NMS keeps the boxes with the highest confidence scores and removes the extra boxes, so the final detections are most accurate.

3. In your own words, what does IoU measure?

IoU measures how close the model's prediction box overlaps compared to the actual object. The more over-lap there is the better the prediction was. A high IoU means the model was pretty accurate, but a low IoU means the prediction was way off.

Part 2: Hands-on experience

4. When you ran YOLO11 on your own image, did it detect everything you expected? Did it detect anything that was wrong? Why do you think it succeeded or failed?

YOLO11 detection on my image was really accurate. It detected the main 3 objects the image, two persons and a tie. I considered it a success because nothing was not detected. There were no errors.

5. When you compared detection and segmentation side by side, name one situation where the box would be enough and one situation where you would really need the mask.

A bounding box would be enough when the goal is knowing what object is present and where. If a construction site wanted to count total workers on a job

site, detection would be enough. One instance where the mask is preferred is where there are multiple objects in a similar vicinity. So, a company needing to know how many workers are on at work and who has their vests on, detection might not be enough because the boxes would overlap but the segmentation can more easily separate the objects.

6. SAM 2 does not output category labels. Why is that an important difference from YOLO?

Its really important because YOLO can identify both what the object is and where it is. But SAM2 is different. It can identify only that its something. Basically object1, object2, object3. SAM2 is better at drawing the mask more accurately and outlining shapes. But YOLO identifies what it is.

Part 3: Trade-offs

7. Confidence threshold trade-off. Give one application where you would want a low threshold (more sensitive) and one application where you would want a high threshold. Explain why.

High threshold would be better if you are looking for specific things. So, less detections but very confident scores. Like if a company needed to look for specific safety violations. Less detections but more reliable. Low threshold would be more useful if you just needed to identify as many objects as possible. Lower confidence but the Human-In-The-Loop can determine what's important and what's false alarms, like security footage.

8. Precision versus recall. For a workplace safety system that detects unsafe behavior on a factory floor, do you want higher precision or higher recall? What about for a system that tags products in a customer's photo upload?

For a workplace safety system, I would want higher recall because it is more important to catch as many safety violations as possible. Missing a real safety issue could lead to someone getting hurt. It makes more sense to me having a few false alarms that can be reviewed than completely miss an unsafe situation. For a product tagging system, I would want higher precision because customrs expect the tags to be accurate. Incorrect tags would make it harder to find products and create a poor user experience.

9. When would you reach for a specialist model like YOLO11-seg, and when would you reach for a foundation model like SAM 2?

I would use a specialist model like YOLO11-seg when I know exactly what objects I am looking for and need the model to identify them. For example, if I were building a system to identify different food items on a plate, I would need

the model to recognize specific categories of food. I would use a foundation model like SAM 2 when I need accurate segmentation of objects but do not necessarily need labels. SAM 2 is useful for separating objects from the background, while YOLO11-seg is better when I need both the object location and the object category.

Part 4: Midterm project

10. What project area did you pick for your midterm? In one paragraph, describe: (a) what the application does, (b) where the data will come from, and (c) how you might extend it for the capstone.

For my midterm project, my group actually chose to work on a Food Recognition and Meal Logging system. The goal of the application is to use image classification to identify common foods from a photo and suggest meal entries instead of requiring users to manually search for every food item. This could make calorie tracking faster and easier for people who regularly log their meals. The data will come from a public food image dataset, such as Food-101, which contains thousands of labeled food images. For the capstone project, we would like to improve the model by testing it on real meal photos, increasing the number of food categories it can recognize, and adding basic calorie or nutrition estimates based on the foods that are identified.